

Quan Vu

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EDUCATION

University of California, San Diego - Computer Engineering, B.S. - 4.00 GPA

Sep 2025 - Present

California Academy of Mathematics and Science - 4.52 GPA (4.0 UW)

Sep 2021 - Jun 2025

EXPERIENCE

CS Club President | California Academy of Mathematics and Science

Aug 2023 - Jun 2025

- Organized coding workshops and lessons to develop club members' skills in Python and Java.
- Coordinated game jams & programming challenges with the computer science teacher, allowing members to practice their programming skills practically and collaboratively.
- Led a 10-person team in the American Computer Science League (ACSL) competition through regular study sessions.

Machine Learning Intern @ VinAI Residency Program

Jun 2024 - Jul 2024

- Studied machine learning under the guidance of VinAI's research scientists & graduate students, learning to develop and implement basic neural networks for recommendation systems and image classification in Pytorch.
- Trained and fine-tuned a neural network & utilized OpenCV to make a self-driving RC car that could navigate a track and recognize stop signs.

Calculus Teacher Assistant / Math Tutor

Jun 2022 - Aug 2024

- Provided in-class support for an AP Calculus BC class during my junior year, assisting peers with class activities, homework assignments, and test corrections.
- Tutored a high school student diagnosed with autism, helping them catch up in algebra by learning to adapt the way I present information and fostering a stress-free environment to accommodate their nuances.

PROJECTS

Accessible Shopping List | React Native · JavaScript · OpenAI API · SerpAPI

[\(Github repo\)](#)

- Developed a mobile application designed to assist individuals with intellectual disabilities in shopping by displaying written items in a shopping list with pictures, along with a product recognition feature using the device camera.
- Integrated OpenAI's **Vision API** for object recognition from photos taken with the camera & **SerpAPI** to fetch relevant product images from Google Images.
- Implemented an intuitive design for accessibility.

Guitar Chord Book | Python · Django · MongoDB · AWS · React JS · HTML · Tailwind CSS

[\(Github repo\)](#)

- Developed a full stack web application that lets users search and save guitar chord diagrams for their favorite songs and chord progressions.
- Built a web scraper that retrieves chord diagram urls in the backend, which get saved into a MongoDB Atlas cluster.
- Deployed the React frontend and Dockerized Django backend on an AWS EC2 Linux instance.

Real-Time Hand Gesture Recognition Model | Pytorch · Python · OpenCV

[\(Github repo\)](#)

- Trained a ResNet18-based gesture classification model in PyTorch using a subset of the HaGRID dataset, implementing bounding-box cropping, data augmentation, and a custom training pipeline in Google Colab.
- Built a real-time webcam demo that detects hands with MediaPipe, crops the region, and classifies gestures using the trained model; developed with Python, PyTorch, OpenCV, and NumPy.

RELEVANT COURSEWORK

Data Structures and Object-Oriented Design, Advanced C++, Linear Algebra, Systems Programming

HOBBIES

Guitar, Tennis, Pickleball, Finding Good Food, Eating, Movies & Shows, Weightlifting, Videogames, Traveling